A 40-year-old PWD store owner rarely uses appliances,
yet pays P1,000+ monthly for electricity.



"Hindi ko naman ginagamit, bakit kailangan ko bayaran?"

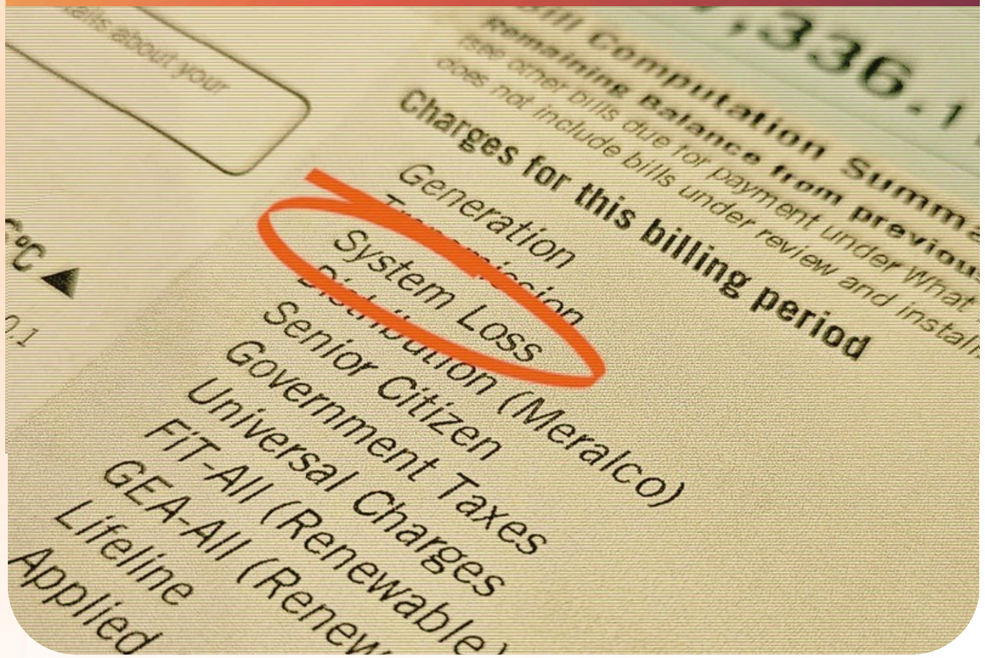
Her question exposed a bigger one:
where is the power going?

SYSTEM LOSS

25 Years

of system-loss charges on Filipino bills

- Inquirer.net (2026)



₱6.8B

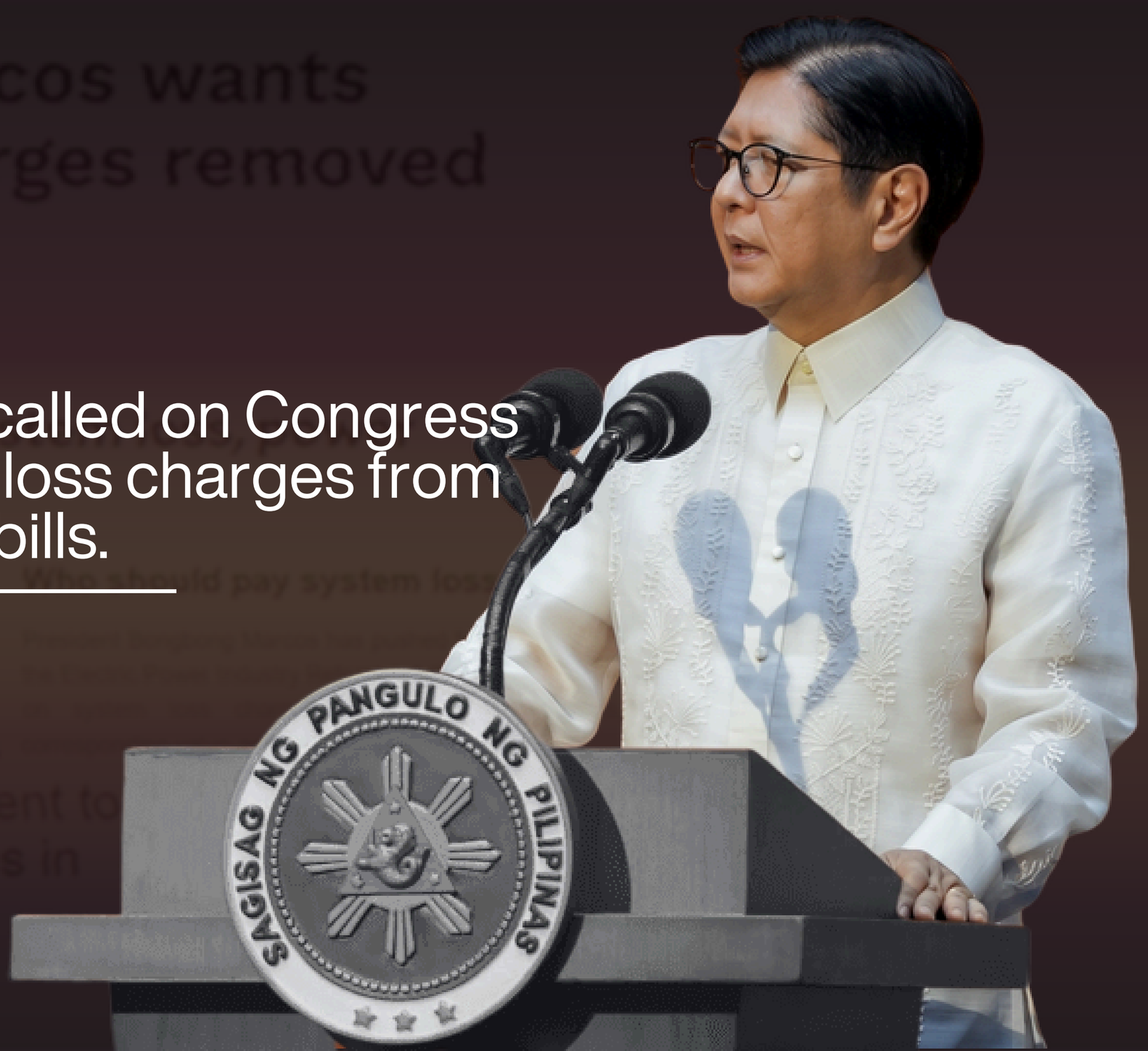
estimated yearly cost tied to electricity theft

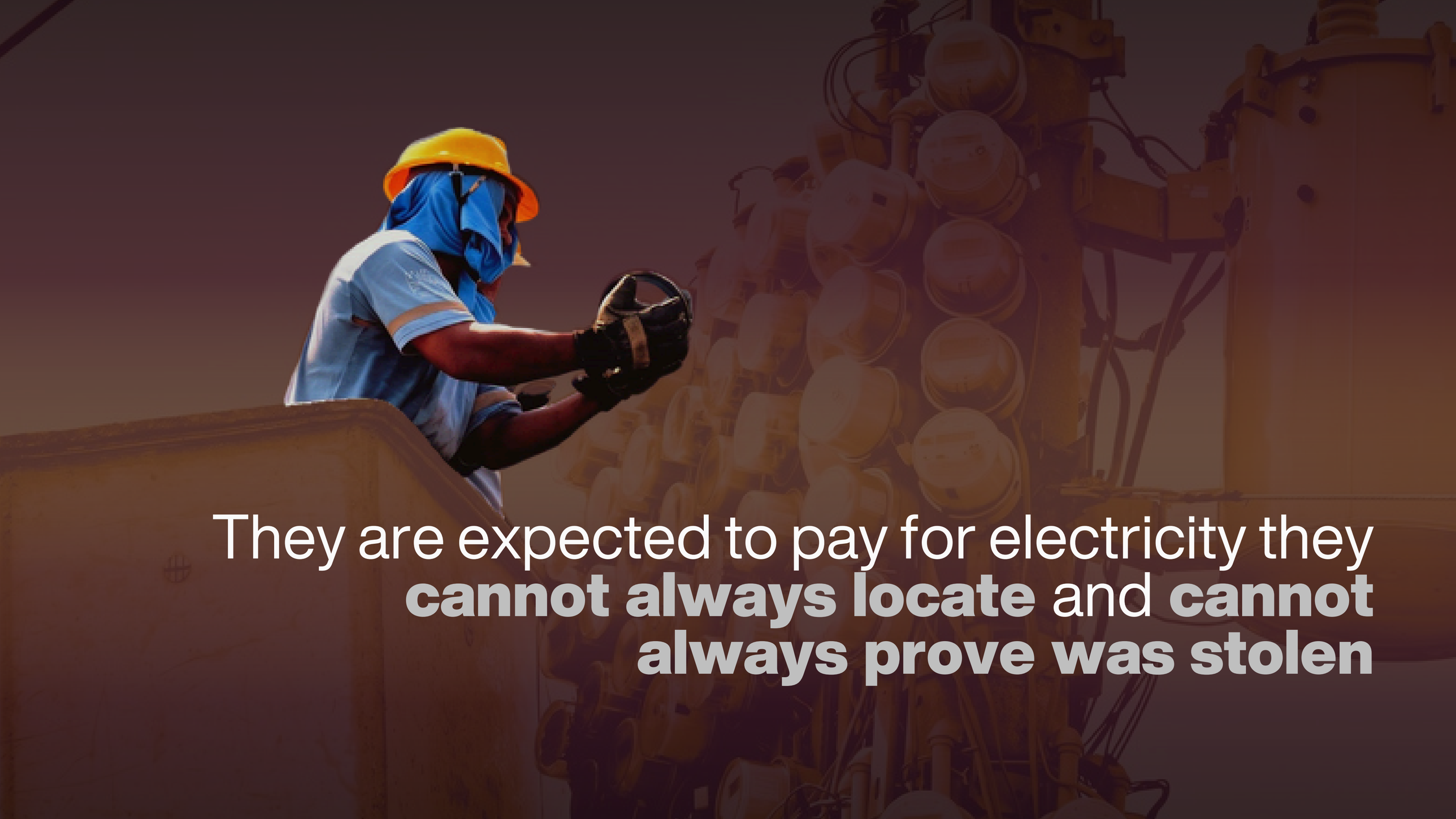
- Daily Tribune (2026)



President Marcos called on Congress to **eliminate** system loss charges from consumer electricity bills.

July 2026





They are expected to pay for electricity they **cannot always locate** and **cannot always prove was stolen**

21/548

pilferage cases since 2022 ended in conviction

469 cases are still pending in court. 314 were filed in Calabarzon, the most of any region.

Ceneco, Negros (2022)

Pays for tips. Inspection crew grew from 3 to 8.

ZAMCELCO, Zamboanga (2023)

Takes tips by call or email. Sends inspection teams.

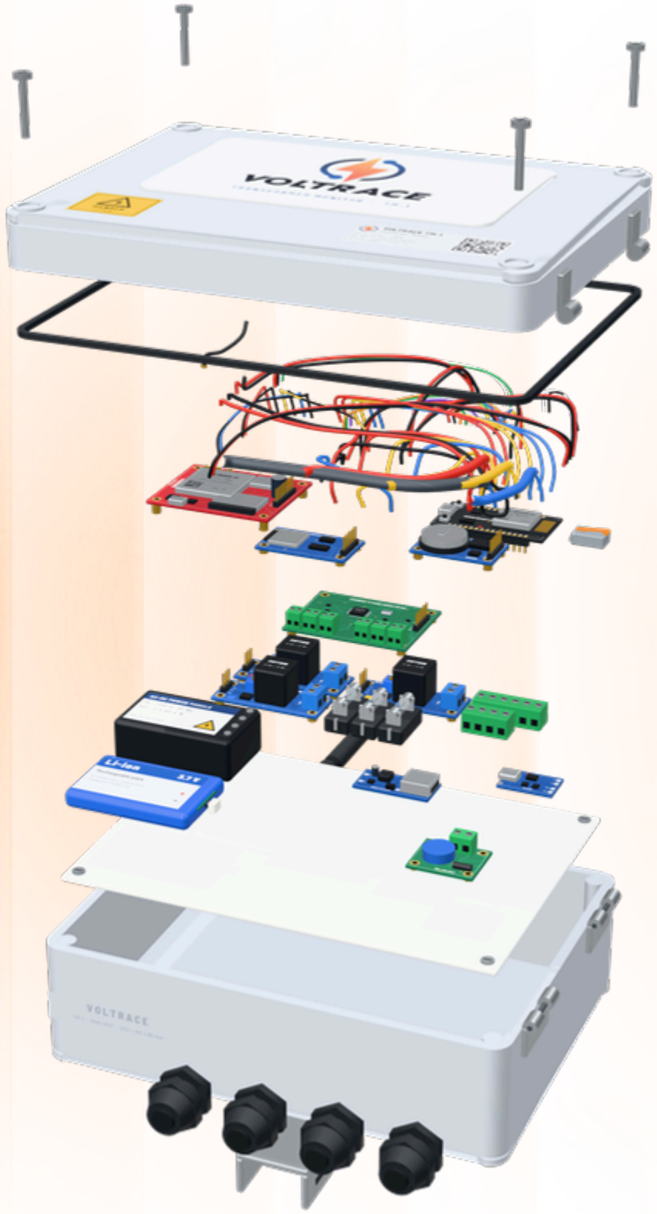

What if **electric cooperatives** didn’t
have to **search blindly?**

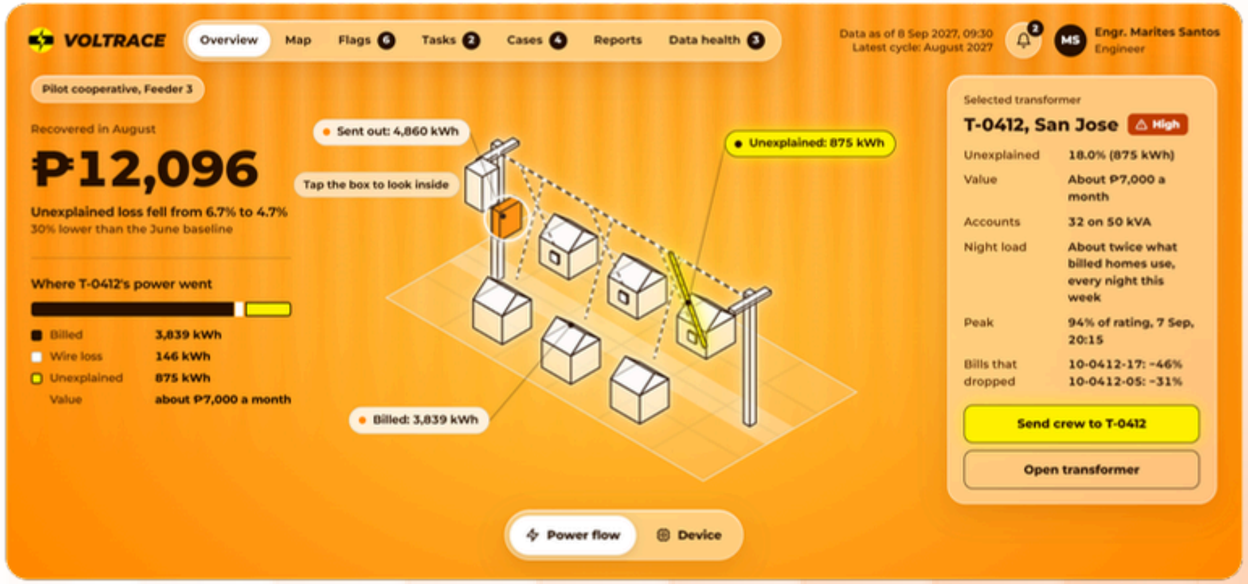
VOLTRACE

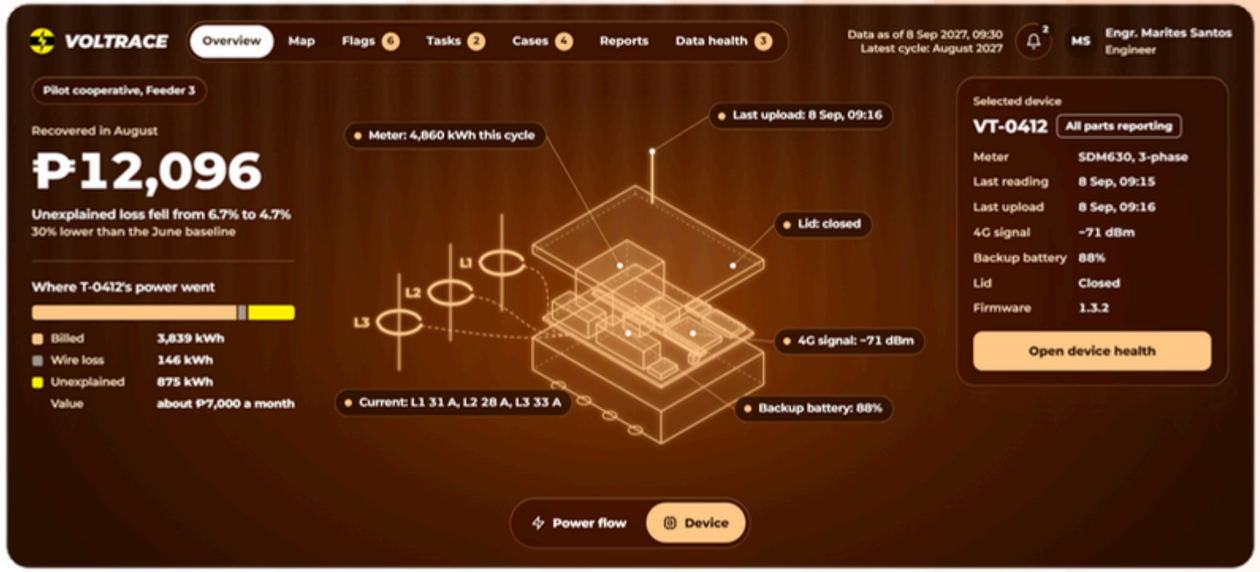
Know where the power goes

An **AI, IoT and SaaS** platform that shows cooperatives which transformers to inspect first.




VOLTRACE


VOLTRACE
Overview
Map
Flags 6
Tasks 2
Cases 4
Reports
Data health 3
Data as of 8 Sep 2027, 09:30
Latest cycle: August 2027
MS
Engr. Marites Santos
Engineer
Pilot cooperative, Feeder 3
Recovered in August
₱12,096
Unexplained loss fell from 6.7% to 4.7%
30% lower than the June baseline
Where T-0412's power went
Billed 3,839 kWh
Wire loss 146 kWh
Unexplained 875 kWh
Value about ₱7,000 a month
Sent out: 4,860 kWh
Unexplained: 875 kWh
Tap the box to look inside
Billed: 3,839 kWh
Selected transformer
T-0412, San Jose
High
Unexplained 18.0% (875 kWh)
Value About ₱7,000 a month
Accounts 32 on 50 kVA
Night load About twice what billed homes use, every night this week
Peak 94% of rating, 7 Sep, 20:15
Bills that dropped 10-0412-17: −46%
10-0412-05: −31%
Send crew to T-0412
Open transformer
Power flow
Device


VOLTRACE
Overview
Map
Flags 6
Tasks 2
Cases 4
Reports
Data health 3
Data as of 8 Sep 2027, 09:30
Latest cycle: August 2027
MS
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Where T-0412's power went
Billed 3,839 kWh
Wire loss 146 kWh
Unexplained 875 kWh
Value about ₱7,000 a month
Meter: 4,860 kWh this cycle
Last upload: 8 Sep, 09:16
Lid: closed
4G signal: −71 dBm
Backup battery: 88%
Current: L1 31 A, L2 28 A, L3 33 A
L1
L2
L3
Selected device
VT-0412
All parts reporting
Meter SDM630, 3-phase
Last reading 8 Sep, 09:15
Last upload 8 Sep, 09:16
4G signal −71 dBm
Backup battery 88%
Lid Closed
Firmware 1.3.2
Open device health
Power flow
Device

VOLTRACE

Measures

Voltage, current, power and energy (kWh)

Connects

LTE / 4G for real-time data transmission

Analyzes

AI detects abnormal patterns and prioritizes areas

Helps

Cooperatives reduce losses, improve efficiency and serves communities better.

Narrowing the search from thousands of homes to 10-50 per transformer.

How the data reaches the software

Measure

CT rings and voltage sensors read the line and the following:

Voltage
Current
Power
Energy (kWh)

Record

The ESP32 saves each reading with the time.

Store

Readings stay on the SD card if there's no signal.

Send

LTE uploads the day's data to the cloud.

Analyze

AI compares it with billing data and ranks where to inspect.

Our focus is the

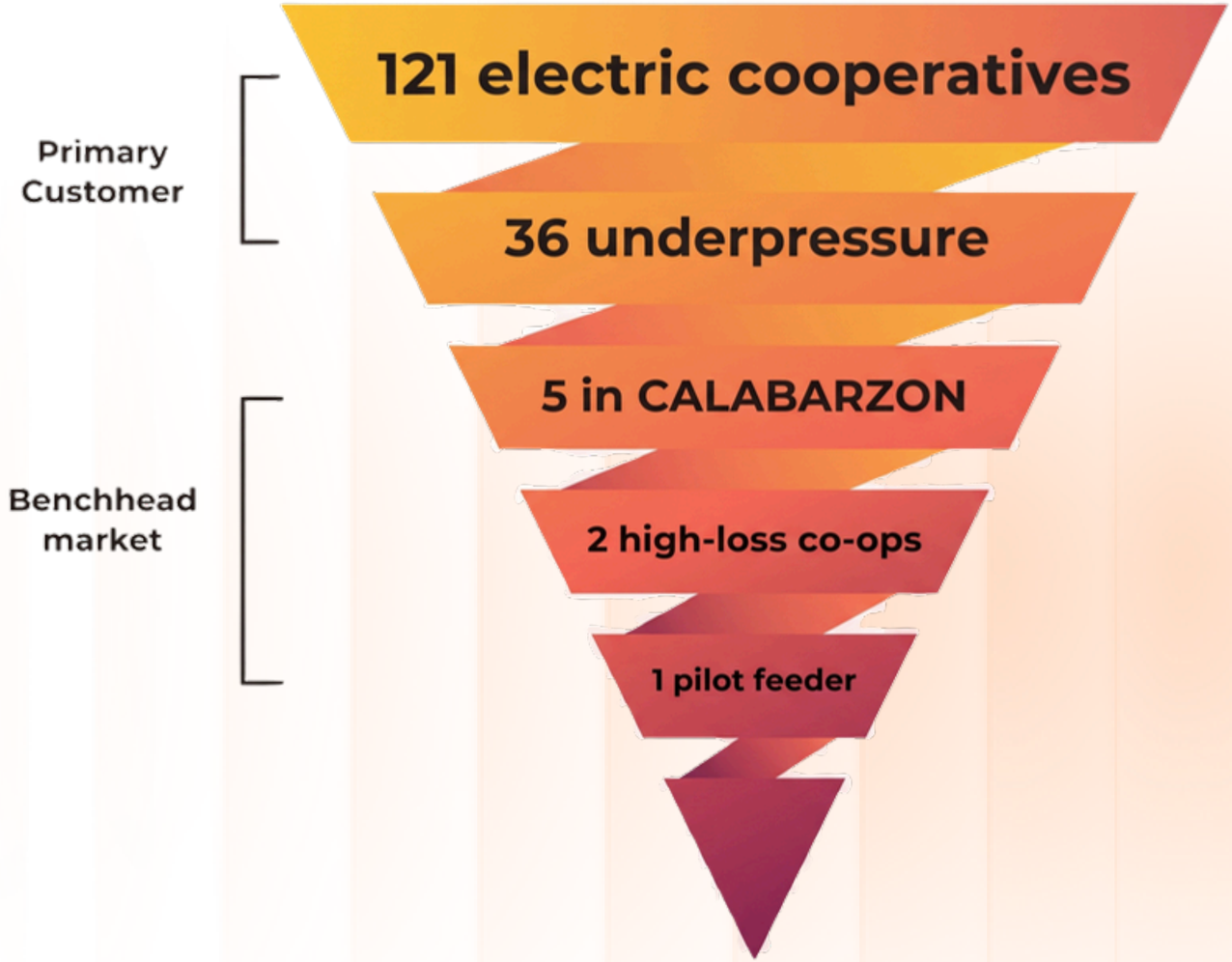
36 cooperatives

that exceeded their Energy Regulatory Commission (ERC) loss caps in May 2026

Beachhead Target

QUEZELCO I and II

Breached ERC system loss caps.
Accessed via ASTEC network.



PROGRESS TRACTION

OCT TO DEC 2026

BUILD

Prototype development
Target: ±2% accuracy

NOV 2026 TO FEB 2027

VALIDATE

3-coop engagement
Target: 90% benchmark detection

MAR TO SEP 2027

PILOT

three feeder
Measure inspection and loss outcomes

OCT 2027–2029

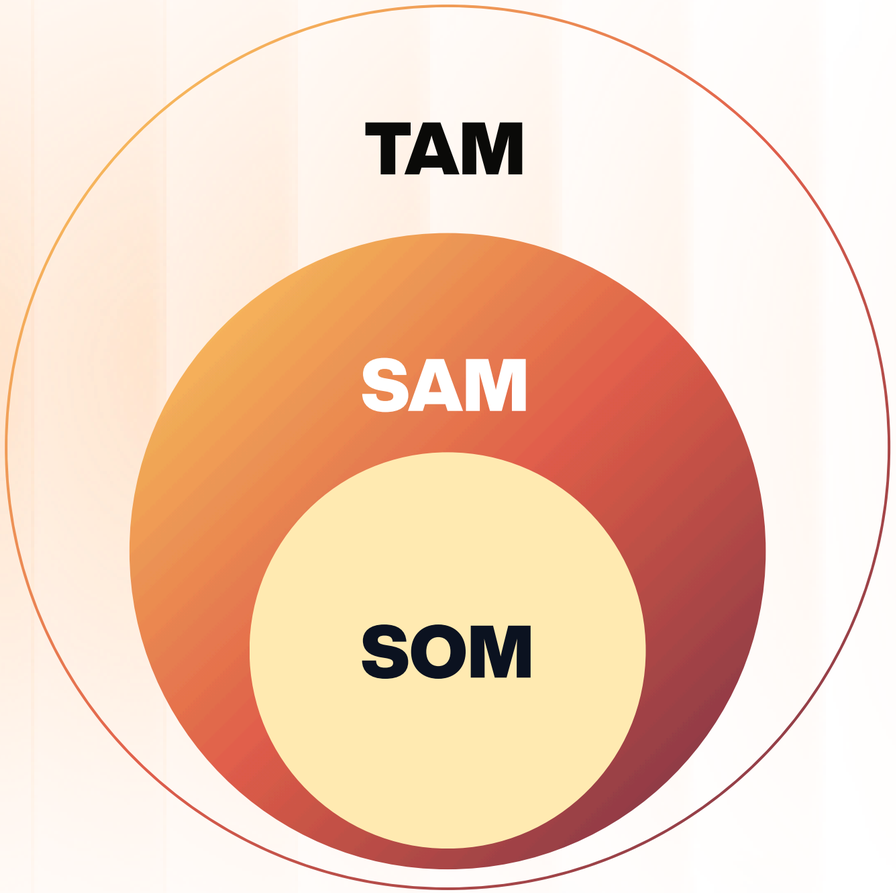
SCALE

Paid contract, certification, 5 coops and 3,000 devices via ASTEC and PHILRECA


2030
EXPAND
Track verified system-loss fixes
Begin technical-loss analysis

	Manual Inspection	Household Smart Meters	SparkMeter (in Philippine pilots)	Global Transformer Monitors	Voltrace
Finds theft by transformer	⃠	Partial	✓	✓	✓
Pinpoints the exact house	Partial	✓	Not advertised	⃠	⃠
No need to replace household meters	✓	⃠	Not advertised	✓	✓
Daily inspection tasks for crews	⃠	Not advertised	Not advertised	Not advertised	✓
Court-ready evidence files	⃠	⃠	Not advertised	Not advertised	✓
Pay-from-recovery option	—	⃠	Not advertised	Not advertised	✓
No new equipment needed	✓	⃠	⃠	⃠	⃠
Proven field deployments	✓	✓	✓	✓	⃠
Certified for utility use	—	✓	✓	✓	⃠ (in progress)

MARKET SIZING



TAM For all **121 electric cooperatives**: ₱1.1 billion in hardware + ₱719 million a year in service (~75,000 units)

SAM Focused on the **₱358 Million** market of 36 high-loss cooperatives that breached ERC caps: ₱335 million in hardware + ₱214 million a year (~22,000 units)

SOM By 2029, 5 cooperatives starting in CALABARZON: ₱45 million in hardware + ₱29 million a year (3,000 units)

HARDWARE–ENABLED SaaS

₱14,999

₱1,305/month for 12 months (8%)

per unit:
device + 3 clamps, installation + 1 - year warranty cash

₱799

per unit per month

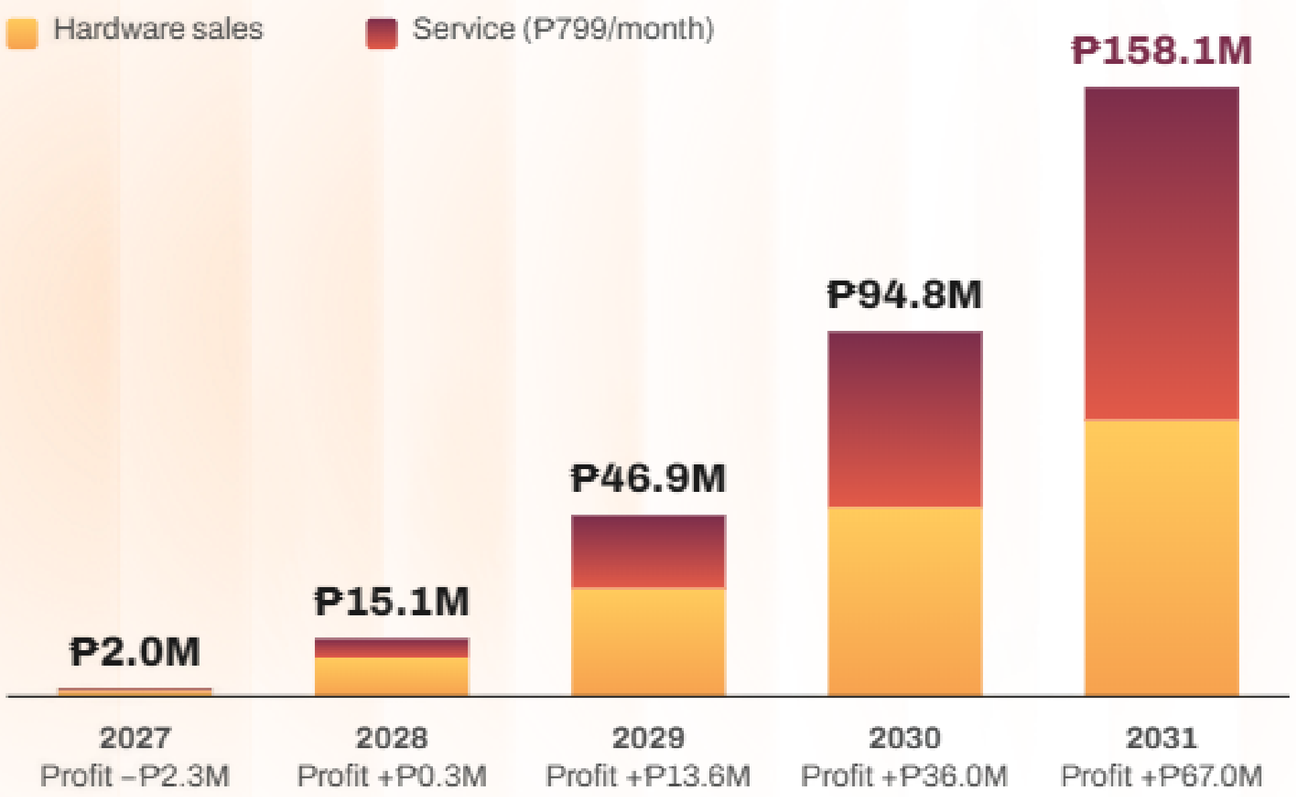
per unit, per month:
software + data + on-site repairs + quarterly relocation+ evidence files

~5 months

payback on a hotspot transformer

REVENUE BY 2031

Assumes ₱14,999 per unit + ₱799/month, 60% cash and 40% installment, unit cost falling to ₱6,500.



11,520
units installed across 16 co-ops

₱110M
yearly service revenue by end of 2031

58%
gross margin in 2031

Year 2
break-even (2028)

₱850,000 to build, validate, and execute a 20-unit cooperative pilot in 12 months

₱850,000 IN GRANTS

From DICT, DOST-PCIEERD and our school incubator.

Builds, tests and pilots Voltrace

IN-KIND SUPPORT

Linemen, site access and billing data.

Field deployment

INVESTOR FUNDING

DTI–NDC Venture Fund and impact and energy investors.

Scale to 3,000 devices

Category	Amount	Share
Equipment: pilot hardware (20 field-grade units at ₱10,000)	₱200,000	24%
Operations: connectivity and cloud (25 devices, 12 months)	₱126,000	15%
Operations: field work (travel, installation support, safety gear)	₱120,000	14%
Staffing: part-time team stipends (4 members, 10 months)	₱100,000	12%
Operations: certification and legal (NTC, data privacy, trademark)	₱100,000	12%
Contingency	₱69,000	8%
Product development: software (tools, maps, app publishing)	₱50,000	6%
Product development: prototypes and testing	₱45,000	5%
Marketing and partnerships (ASTEC and PHILRECA engagement)	₱40,000	4%
Total	**₱850,000**	**100%**

SDG 11



Affordable and
Clean Energy

SDG 9



Industry, Innovation
and Infrastracture

Behind every line on a bill is a person asking, “bakit?”

Meet the team that decided to answer.



MHALIK B. PEREZ

Hack4aprogress (Top 3 Finalist)
CTI SUCCESS 2026



JOHN VINCENT M. DORIA

DOST STTP (CALABARZON Representatives)
Hack4Aprogress (Top 3 Finalist)
CTI SUCCESS 2026
Industry Innovation 2026
Globe Launchpad 2026



SOFHIA AUBREY M. ASILO

CTI SUCCESS 2026
Globe Launchpad 2026



STARTUP MENTOR

GLENN M. GUINOBAN

Lecturer I - BatStateU Alangilan
Robotics Solutions Specialist & Founder - Robolicious Philippines

VOLTRACE

Contact Information

Scan Me!

Web Application

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voltrace.jumpers@gmail.com

https://www.facebook.com/profile.phpid=61595427607254